King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 102 Major Exam I Term 243

Original exam date: 3 July 2025. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. $\int_{-3}^{3} \left(x^3 + 4x^2 - 3x - 6\right) dx =$

(a) 9

(b) 16

(c) 25

(d) 49

(e) 36

2. $\lim_{n\to\infty} \frac{1}{n^3} \left(1^2 + 2^2 + 3^2 + \cdots + n^2\right) =$

(a) $\frac{1}{2}$

(b) $\frac{1}{3}$

(c) $\frac{1}{5}$

(d) $\frac{1}{6}$

(e) $\frac{1}{4}$

3. The area of the region bounded by the graphs of the functions $f(x) = \sin x$ and $g(x) = \cos x$ over $[0, \pi]$ is equal to

(a) $\sqrt{2}$

(b) $2\sqrt{2}$

(c) $3\sqrt{2}$

(d) $2 + 2\sqrt{2}$

(e) $2 - \sqrt{2}$

4. $\int \frac{\sinh x}{1+\sinh^2 x}\,dx =$

(a) $\cosh x + C$

(b) $\operatorname{sech} x + C$

(c) $\operatorname{csch} x + C$

(d) $-\operatorname{sech} x + C$

(e) $-\operatorname{csch} x + C$

5. The area of the region bounded by the graphs of the functions $f(x) = x^2 + 2x$ and $g(x) = x + 2$ is equal to

(a) $\frac{9}{2}$

(b) $\frac{11}{2}$

(c) $\frac{7}{2}$

(d) $\frac{3}{2}$

(e) $\frac{5}{2}$

6. $\int e^{-x} \tan\left(e^{-x}\right) dx =$

(a) $\ln|\csc(e^{-x})| + C$

(b) $\ln|\sin(e^{-x})| + C$

(c) $\ln|\cos(e^{-x})| + C$

(d) $\ln|\sec(e^{-x})| + C$

(e) $\ln|\tan(e^{-x})| + C$

7. The average value of the function $f(x) = \dfrac{4(x^2+1)}{x^2}$ over $[1, 3]$ is equal to:

(a) $\frac{13}{3}$

(b) $\frac{14}{3}$

(c) $\frac{11}{3}$

(d) $\frac{17}{3}$

(e) $\frac{16}{3}$

8. The upper sum (in terms of n, the number of subintervals) for the region bounded by the graph of the function $f(x) = 7x^2$ and the x-axis on the interval $[0, 3]$ is:

(a) $63 - \dfrac{189}{2n} + \dfrac{63}{2n^2}$

(b) $63 + \dfrac{189}{2n} + \dfrac{63}{2n^2}$

(c) $189 + \dfrac{63}{2n} + \dfrac{63}{2n^2}$

(d) $189 - \dfrac{63}{2n} - \dfrac{63}{2n^2}$

(e) $63 + \dfrac{189}{2n} - \dfrac{63}{2n^2}$

9. Use the midpoint and 4 rectangles to find an approximation of the area of the region bounded by the graph of the function $f(x) = 16 - x^2$ and the x-axis over the interval $[0, 4]$, then the approximated area is

(a) 41

(b) 45

(c) 49

(d) 43

(e) 47

10. $\int \frac{2x^2+7x-3}{x-2}\, dx =$

(a) $x^2 - 11x + 19 \ln|x - 2| + C$

(b) $x^2 + 11x + 19 \ln|x - 2| + C$

(c) $x^2 - 19 \ln|x - 2| + C$

(d) $x^2 + 11x + C$

(e) $x^2 + 11x - 19 \ln|x - 2| + C$

11. $\int_0^2 [\![x]\!]\, dx =$ (Hint: find the area under the curve over $[0, 2]$)

(a) 2

(b) $\dfrac{1}{2}$

(c) 1

(d) 3

(e) $\dfrac{3}{2}$

12. $\int \frac{2\,dx}{\sqrt{-x^2+4x}} =$

(a) $2\arcsin\left(\dfrac{x-2}{2}\right)+C$

(b) $\arcsin\left(\dfrac{x-2}{2}\right)+C$

(c) $\arcsin(x-2)+C$

(d) $2\arcsin(x-2)+C$

(e) $2\arccos\left(\dfrac{x-2}{2}\right)+C$

13. If $F(x)=\int_0^{2x}\cos t^4\,dt$, then $F'\left(\dfrac{\sqrt[4]{\pi/3}}{2}\right)=$

(a) 4

(b) 2

(c) 16

(d) 8

(e) 1

14. Let the velocity function $v(t)=\cos t$ in feet per second, be given for a particle moving along a straight line, where t is the time in seconds. The total distance (in feet) that the particle travels over the interval $[0,2\pi]$ is equal to

(a) 4

(b) 1

(c) 2

(d) 0

(e) 3

Answer Key

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Q	Ans	Brief explanation
1	**(e)**	Odd terms vanish; $2\int_0^3(4x^2-6)dx = 2(36-18) = 36$.
2	**(b)**	$\frac{(n+1)(2n+1)}{6n^2} \to \frac{1}{3}$ (also $\int_0^1 x^2dx = \frac{1}{3}$).
3	**(b)**	Split at $\frac{\pi}{4}$: $(\sqrt{2}-1)+(1+\sqrt{2}) = 2\sqrt{2}$.
4	**(d)**	$1+\sinh^2 x = \cosh^2 x$, integrand $= \operatorname{sech} x \tanh x$, so $-\operatorname{sech} x + C$.
5	**(a)**	Meet at $x=-2,1$; $\int_{-2}^1(2-x-x^2)dx = \frac{9}{2}$.
6	**(c)**	$u = e^{-x}$: $-\int \tan u\, du = \ln\lvert\cos(e^{-x})\rvert + C$.
7	**(e)**	$\frac{1}{2}\int_1^3(4+4x^{-2})dx = \frac{1}{2}\cdot\frac{32}{3} = \frac{16}{3}$.
8	**(b)**	Right endpoints, $\Delta x = \frac{3}{n}$: $\frac{189}{n^3}\cdot\frac{n(n+1)(2n+1)}{6} = 63 + \frac{189}{2n} + \frac{63}{2n^2}$.
9	**(d)**	$\Delta x = 1$, midpoints $\frac{1}{2},\frac{3}{2},\frac{5}{2},\frac{7}{2}$: $\frac{63+55+39+15}{4} = 43$.
10	**(b)**	Division: $2x+11+\frac{19}{x-2}$, so $x^2+11x+19\ln\lvert x-2\rvert + C$.
11	**(c)**	Step function: 0 on $[0,1)$, 1 on $[1,2)$, so the area is 1.
12	**(a)**	$-x^2+4x = 4-(x-2)^2$: $2\arcsin\frac{x-2}{2} + C$.
13	**(e)**	$F'(x) = 2\cos(16x^4)$; $16x^4 = \frac{\pi}{3}$, so $F' = 2\cdot\frac{1}{2} = 1$.
14	**(a)**	$\int_0^{2\pi}\lvert\cos t\rvert dt = 4\int_0^{\pi/2}\cos t\, dt = 4$.